

Sales Data Cleaning & Analysis Report

Transforming Legacy Operational Records into Reliable Business Intelligence

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Date: June 2026

Project Overview

This project focuses on cleaning, validating, and analyzing a sales dataset containing over 2,000 records. The dataset included various data quality issues that could have impacted reporting accuracy and business insights. In addition to correcting historical data, a structured workflow was developed to help prevent similar issues in future datasets.

The Data Preparation & Cleaning Architecture

1. Systematic Assessment

Before analysis began, the dataset was examined for common data quality problems such as missing values, duplicate records, placeholder text errors, and inconsistent data formatting.

2. Scripted Transformation Pipeline

Rather than applying static fixes, transformations were formalized into automated scripts using **Power Query** to establish a repeatable ETL routine:

- **Wildcard Resolution:** Escaped literal text errors (such as strings of ???) using structured search patterns (~?~?~?) to safeguard surrounding string rows from bulk-deletion tools.
- **Deduplication:** Duplicate records were identified and removed using unique identifiers, ensuring the dataset accurately reflected actual sales activity.
- **Missing Value Strategies:** Missing numerical values were identified and handled using Power Query and targeted bulk-fill techniques, ensuring calculations remained accurate and reliable.
- **Formatting Uniformity:** Enforced case-matching standards across inconsistent attributes and enforced rigorous column casting into designated text, currency, and date formats.

3. Implementation of Preventative Quality Controls

To prevent re-contamination by downstream data entry teams, a series of preventative system restrictions were built directly into the tracking sheet:

- **Data Validation Lists & Named Ranges:** Replaced free-text input fields with dropdown lists driven by dynamic Named Ranges to completely eliminate region typos.

- **Passive Governance:** Isolated structural setting arrays into discrete backend configuration tabs. These sheets were set to a hidden system visibility state via the VBA compiler to protect formulas from unauthorized manual adjustments.

System Note: Consolidating these data transformation steps into automated Power Query steps ensures that newly appended rows are instantly cleaned upon a unified dataset refresh.

Key Business Insights

Following the execution of the cleaning pipeline, the structured data yielded clear, unbiased metrics reflecting organizational performance:

TOTAL VALIDATED SALES	TOTAL UNIQUE ORDERS	AVERAGE ORDER VALUE
\$324,400	1,417	\$228.93

Analytical Domain	Core Finding / Highlight	Strategic Action Item
Product Analysis	Widget B registered peak sales revenue across core portfolios.	Optimize supply chain parameters to maintain inventory safety stock.
Regional Variance	The East Region secured the dominant market performance share.	Isolate state-level variables to clone operational models elsewhere.
Order Pipeline	Completed orders drove the major share of financial returns.	Monitor pending funnel statuses to identify logistic bottlenecks.

Interactive Analytics Interface

The cleaned and validated dataset was used to create an interactive dashboard that enables users to explore key business metrics through dynamic filtering and segmentation. The dashboard includes KPI summaries, product performance analysis, and regional insights, providing a comprehensive view of sales performance.

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